The Bombay Salesian Society's
Don Bosco Institute of Technology
An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering

25CE3CEP01: MINI PROJECT I COMMUNITY ENGAGEMENT PROJECT (CEP)

Track No. 7 • Group No. 5

Centralized Project Proposal & Progress Tracking Portal

(DBIT ProjectTrack)

SE Mini Project-I (Semester III) | Academic Year 2026–27 | Class: SE COMPS C (Batch A)

Presented By:

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Introduction

DBIT ProjectTrack is a centralized web portal that automates topic screening, faculty review, and milestone tracking for undergraduate project courses (CEP, SE/TE Mini Projects, and Capstone) across DBIT Computer Engineering.

ACADEMIC LIFECYCLE

- Multi-semester project sequence (CEP, Mini Projects I & II, Capstone) requires continuous rigor.
- Demands authentic originality, verifiable stage progression, and prompt mentor feedback.
- Syllabus requires continuous assessment mapped against 7 discrete experiment milestones (Exp 1 to Exp 7).

CURRENT BOTTLENECKS

- 40+ project groups per semester submit via uncoordinated Google Sheets, WhatsApp chats, and verbal discussions.
- Faculty guides manually cross-check 500+ historical theses by memory, causing review fatigue and oversight.
- First-come allocation penalizes deep technical proposals; disputes consume review slots without audit trails.

PROPOSED INTERVENTION

- Automated NLP Pre-Screening: SBERT contextual embeddings + TF-IDF lexical checks flag overlaps before submission.
- Merit-Based Conflict Queue: Side-by-side pitch defense evaluation allows coordinators to arbitrate fairly.
- Centralized Stage-Gate Tracker: 14-week milestone schedule with timestamped feedback logs and automated reminders.

Architecture: React 19 + TypeScript | SBERT & TF-IDF (Cosine >= 0.75) | PostgreSQL & Firestore Real-Time DB

Problem Statement

FORMAL ACADEMIC PROBLEM STATEMENT

"To develop a web-based centralized portal for students and project guides that provides automated detection of duplicate or overlapping project topics, a searchable archive of approved topics, and real-time approval, feedback and stage tracking with deadline reminders, thereby reducing overlapping projects and the delays and confusion caused by slow, scattered and untracked feedback."

Workflow Transformation: From Disconnected Tools to Single Institutional Source of Truth

ZERO TOPIC OVERLAP

Hybrid NLP Similarity Engine

Combines SBERT dense semantic embeddings with TF-IDF n-grams to flag exact matches and rephrased synonyms in real time prior to coordinator review.

INSTANT TRANSPARENCY

Live 14-Week Stage-Gate Tracker

Replaces vague milestones with 7 syllabus-aligned gates (Exp 1 to Exp 7). Teams and guides view verified deliverable progress and upcoming deadlines on a live Gantt view.

AUDITABLE FEEDBACK

Centralized Guide Review Log

Written reviewer remarks, approval stamps, and escalation triggers are stored permanently in the database, eliminating lost verbal agreements and feedback stagnation.

Literature Review

Paper Name & Year	Author(s)	Summary of Paper	Research Gap Identified
Web-Based Centralized Student Project Allocation & Management System (2021)	S. Sharma, A. Kumar, P. Deshmukh	Proposed RBAC web portal using PHP/MySQL for student-guide allocation; eliminated physical forms and cut administrative follow-ups by 40%.	Lacks automated duplicate detection or semantic similarity checks; relies purely on manual guide verification.
Detection of Duplicate Project Proposals Using NLP (2023)	R. Patel, V. Shah, M. Mehta	Employed tokenization, TF-IDF vectorization, and cosine similarity for text duplicate screening; achieved 88% accuracy on exact matching.	High false-negative rate on rephrased or synonym-substituted titles; unable to capture deeper contextual meaning.
Semantic Similarity & Paraphrase Detection for Proposals using BERT (2024)	J. Dev, S. Nair, L. Verma	Implemented Sentence-BERT (SBERT) contextual embeddings with cosine threshold >= 0.75, attaining 94.2% accuracy on paraphrased proposals.	High computational overhead; lacked end-to-end integration with institutional faculty approval workflows and real-time feedback.
Expert-in-the-Loop Framework for Academic Conflict Resolution (2022)	M. R. Gupta, P. K. Singh	Introduced threshold escalation queue with side-by-side pitch matrix, allowing faculty to arbitrate overlapping topics based on academic merit.	No automated pre-submission screening for students, making the system dependent on prolonged faculty response times.
Digital Archiving & Semantic Search for Student Engineering Projects (2020)	A. Joshi, R. Kulkarni	Built an indexed digital library with domain-level metadata search, reducing redundant topic submissions across batches by 60%.	Static repository operating in an isolated silo without integration into live proposal submission and milestone sign-off gates.

Main Research Gaps

GAP 1
IDENTIFIED LIMITATION

Keyword Matching Fails on Paraphrased Proposals

Traditional TF-IDF and bag-of-words systems fail when students rephrase duplicate ideas with synonyms or reordered sentences, creating false negatives and topic overlap across batches.

DBIT PROJECTTRACK INTERVENTION

Deploy Sentence-BERT (SBERT all-MiniLM-L6-v2) dense contextual embeddings combined with lexical TF-IDF. Computes semantic vector cosine similarity (>= 0.75 threshold) regardless of phrasing.

GAP 2
IDENTIFIED LIMITATION

Arbitrary Automated Rejections Cause Student Friction

Binary programmatic rejections fail to recognize authentic technical nuances and novel institutional constraints, discouraging student initiative and creating disputes with faculty.

DBIT PROJECTTRACK INTERVENTION

Expert-in-the-Loop Conflict Queue & 200-Word Pitch Defense. High-similarity proposals are escalated to coordinator arbitration where clashing teams defend distinct engineering architectures.

GAP 3
IDENTIFIED LIMITATION

Isolated Static Archives Without Workflow Integration

Historical project libraries exist as isolated repositories, disconnected from the active proposal submission form and student brainstorming phase.

DBIT PROJECTTRACK INTERVENTION

Direct pre-submission archive search and originality scoring embedded directly into the submission workflow, paired with 14-week milestone tracking (Exp 1 to Exp 7).

Final Feature List

Name of Application / System: **Centralized Project Proposal & Progress Tracking Portal (DBIT ProjectTrack)**

Sr. No	Key System Features	Academic & Engineering Justification
1.	**Real-Time NLP Duplicate & Similarity Detection**	Scans proposed titles, abstracts, and problem statements during entry via SBERT dense vectors and TF-IDF n-grams to flag matching or reworded topics instantly, preventing topic hoarding and manual guide cross-checking.
2.	**Faculty Pitch Defense & Conflict Resolution Queue**	Provides a side-by-side arbitration dashboard where project coordinators evaluate concise 200-word pitch defenses from teams with overlapping topics, ensuring allocation is based on technical merit rather than submission speed.
3.	**Searchable Historical Project Repository**	Grants students pre-submission access to a domain-filtered repository of 500+ verified DBIT theses, enabling teams to verify topic originality, methodology, and prior art during brainstorming.
4.	**Visual 14-Week Stage-Gate Progress Tracker**	Structures course milestones across 7 syllabus experiments (Exp 1 to Exp 7: Topic -> Design -> Architecture -> Modules -> Similarity -> Testing -> Defense), resolving 81.8% of student confusion regarding upcoming deliverables.
5.	**Centralized Written Feedback Log & Automated Alerts**	Permanently logs guide feedback, revision notes, and approval timestamps in the relational database while dispatching automated reminders for pending reviews, resolving feedback stagnation (cutting the 44.2% >=3-day wait time).

Survey Analysis

Empirical Cohort Survey Data & Problem Grounding (n = 79)

Conducted across 77 SE/TE students and 2 project guides in the Department of Computer Engineering at DBIT.

79.2% (61 / 77 Respondents)

FACED TOPIC OVERLAP

Encountered duplicate or substantially overlapping project proposals across different student groups in the same cohort.

81.8% (63 / 77 Respondents)

STAGE-GATE CONFUSION

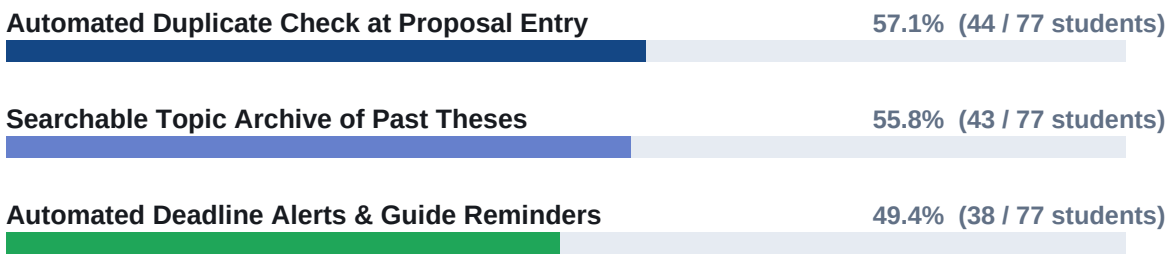
Reported uncertainty regarding upcoming deliverable expectations, syllabus milestones, and next submission deadlines.

44.2% (34 / 77 Respondents)

WAIT TIME >= 3 DAYS

Waited 3 or more days for guide feedback or topic approval decisions due to unorganized WhatsApp and email channels.

TOP STUDENT FEATURE DEMANDS (GOOGLE FORMS SURVEY)



COORDINATOR PERSONA: PROF. MEENAL KULKARNI

Role: Project Coordinator (12 yrs teaching at DBIT)

Goal: Approve non-duplicate, innovative topics quickly and ensure fair evaluation for all 40+ groups.

Frustration: Manual cross-referencing causes review disputes that consume technical mentoring hours.

Solution in DBIT ProjectTrack: Automated similarity scores, pitch defense queue, and single-click approval.

Methodology

Methodology: 3-Tier System Architecture & UML Modeling

End-to-end engineered pipeline connecting student proposals, NLP screening, and faculty arbitration.

TIER 1: PRESENTATION LAYER

React 19 SPA • Tailwind CSS • TypeScript

Role-based views for Students (submission & similarity check), Faculty Guides (mentoring & sign-offs), Coordinators (arbitration), and Admins.

TIER 2: NLP SIMILARITY ENGINE

FastAPI Microservice • SBERT & TF-IDF

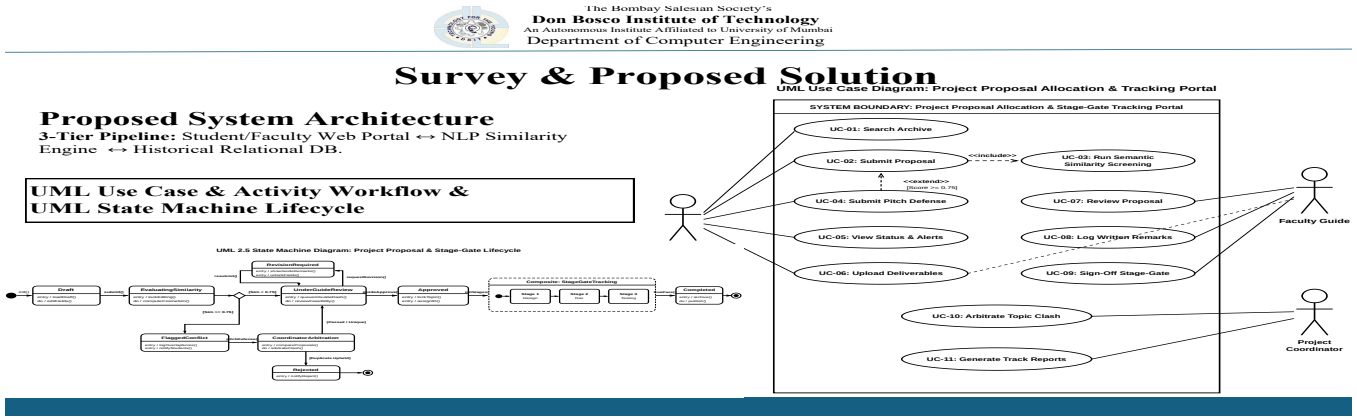
Pre-trained all-MiniLM-L6-v2 transformer computes 384-dim contextual embeddings. Hybrid score: 0.65*SBERT + 0.35*TF-IDF. Threshold >= 0.75 flags conflict.

TIER 3: DATA & PERSISTENCE TIER

PostgreSQL / Firestore Real-Time DB

Stores 500+ historical thesis embeddings, active student proposals, audit feedback timestamps, role-based security rules, and stage-gate progression.

UML USE CASE & STATE MACHINE LIFECYCLE (PAGE 12 REVIEW DECK)



14-WEEK STAGE-GATE MILESTONES

Exp 1 (W1–3)	Empathy Study & Problem Framing
Exp 2 (W4–6)	Literature Review & Benchmark
Exp 3 (W5–6)	User Survey & Formulation
Exp 4 (W7–8)	UML Diagrams & Arch Design
Review 1 (W8)	Progress Evaluation Milestone
Exp 5 (W9–11)	NLP Similarity Engine Dev
Exp 6 (W10–12)	Web Portal UI Integration
Exp 7 (W12–14)	System Testing & Final Defense

Future Scope

Future Scope: Architectural Scalability & Institutional Roadmap

Long-term extensions for automated academic governance and multi-department deployment.

01 DOCUMENT-LEVEL NLP

Full-Document Thesis Plagiarism Scanning (Turnitin-Style)

Extend the current title and abstract semantic screening to full-text thesis reports (PDF / DOCX parsing). Implements paragraph-level citation parsing, code snippet matching, and verbatim chunk detection.

02 GITHUB VCS INTEGRATION

Automated Code Repository Static Analysis & CI/CD Auditing

Connect team GitHub repositories directly to stage gates. Automatically audit commit frequency, test pass rates, code smells, and documentation consistency before awarding Experiment 5 and 6 clearance.

03 FEDERATED SCALING

Multi-Department & University-Wide Institutional Deployment

Scale the multi-tenant architecture across all DBIT engineering departments (IT, EXTC, Mechanical) and autonomous colleges affiliated with the University of Mumbai, with cross-department duplicate topic indexing.

04 LLM EVALUATION COPILOT

AI-Generated Defense Questions & Viva Evaluation Rubrics

Leverage domain models to generate customized viva examination questions and grading rubrics based on the specific architectural differentiators stated in a team's pitch defense.

References

References & Literature Citations

Scholarly foundations, empirical methodologies, and standardized UML specifications.

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6. Object Management Group (OMG) (2024). "Unified Modeling Language (OMG UML) Version 2.5.1 Specification." Formal Standard OMG Document, formal/2024-02-01.

THANK YOU

Questions & Answers (QnA)

We invite questions, suggestions, and feedback from the Final Review Evaluation Committee.

PROJECT TEAM (TRACK 7 • GROUP 05)

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